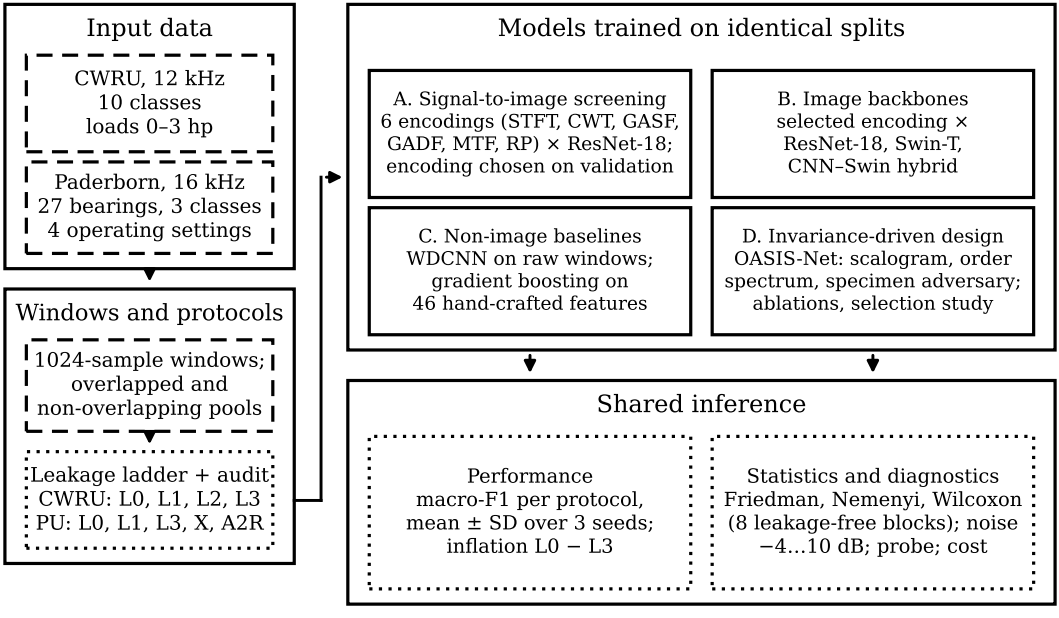
Input data
CWRU, 12 kHz
10 classes
loads 0–3 hp
Paderborn, 16 kHz
27 bearings, 3 classes
4 operating settings
Windows and protocols
1024-sample windows;
overlapped and
non-overlapping pools
Leakage ladder + audit
CWRU: L0, L1, L2, L3
PU: L0, L1, L3, X, A2R
Models trained on identical splits
A. Signal-to-image screening
6 encodings (STFT, CWT, GASF,
GADF, MTF, RP) × ResNet-18;
encoding chosen on validation
B. Image backbones
selected encoding ×
ResNet-18, Swin-T,
CNN–Swin hybrid
C. Non-image baselines
WDCNN on raw windows;
gradient boosting on
46 hand-crafted features
D. Invariance-driven design
OASIS-Net: scalogram, order
spectrum, specimen adversary;
ablations, selection study
Shared inference
Performance
macro-F1 per protocol,
mean ± SD over 3 seeds;
inflation L0 − L3
Statistics and diagnostics
Friedman, Nemenyi, Wilcoxon
(8 leakage-free blocks); noise
−4...10 dB; probe; cost
Data
Model
Analysis